GOVERNMENT ENGINEERING COLLEGE, SIWAN
Mid-semester Examination-2025
B. Tech 1st Semester, CSE(IOT)
Subject: Mathematics-I

Time: 1:30 hrs

M.M: 20 Marks

Instructions:

(i) The marks are indicated in the right hand margin.
(ii) There are **SIX** questions in this paper.
(iii) Attempt **FIVE** questions in all.
(iv) Question no. **1** is **compulsory**.

Q.1: Choose the correct answer: (4)

a) If $AA^T = AA^T = I$ then matrix A is

(i) Symmetric (ii) Skew symmetric (iii) Orthogonal (iv) Idempotent

b) Determinant of the matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ is

(i) 2 (ii) 4 (iii) -2 (iv) -4

c) If the matrix $\begin{bmatrix} 2 & 3 \\ 4 & x \end{bmatrix}$ is singular then x is

(i) 6 (ii) -6 (iii) 3 (iv) 4

d) Let $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} x & y \\ y & x \end{bmatrix}$ then

(i) $x = a^2 + b^2, y = a^2 - b^2$ (ii) $x = a^2 + b^2, y = 2ab$

(iii) $x = a^2 + b^2, y = ab$ (iv) $x = 2ab, y = a^2 + b^2$

Q.2: Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$ by reducing it into normal form. (4)

Q.3: For what value of a and b, the equations (4)

$3x - 2y + z = b,\ 5x - 8y + 9z = 3,\ 2x + y + az = -1$ have unique solutions.

Q.4: Solve the system of equation by Gauss elimination method

$x - y + z = 3,\ x + 2y + 2z = 6,\ x + 5y + 3z = 9$ (4)

Q.5: Find the inverse of the matrix $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$ (4)

Q.6: Let $T: R^3 \to R^2$ be defined by $T(x, y, z) = (x + y, -x + z)$. Show that T is a linear transformation. Also find Kernal of T. (4)